

Proof of conjecture 00000002450: a nondiscrete Polish group all of whose Borel actions are smooth

Jackmeson1

2026-10-05

1 The conjecture and its reading

Statement (English, verbatim). Conjecture: There exists a Polish group G all of whose Borel-action orbit equivalence relations are smooth, while G is nondiscrete (an all-smooth group).

Reading. This is a plain existential statement. We read every term in its standard sense.

- A *Polish group* is a topological group whose topology is Polish (separable and completely metrizable). Lean: `Group G, TopologicalSpace G, IsTopologicalGroup G, PolishSpace G`. Its measurable structure is the Borel σ -algebra (`MeasurableSpace G` together with `BorelSpace G`).
- A *Borel action* of G is a group action $G \times X \rightarrow X$ on a standard Borel space X (`StandardBorelSpace X`) such that the action map $(g, x) \mapsto g \cdot x$ is Borel for the product σ -algebra. Lean: `MulAction G X` together with `Measurable (fun p : G × X => p.1 • p.2)`.
- The *orbit equivalence relation* is $x E_G y \iff \exists g \in G, g \cdot x = y$ (Lean: `orbitEquivRel`).
- An equivalence relation E on X is *smooth* if there are a Polish space Y and a Borel map $f : X \rightarrow Y$ with $x E y \iff f(x) = f(y)$ for all x, y (Lean: `IsSmooth`; Y ranges over Polish spaces in `Type`, with their Borel σ -algebra).
- G is *all-smooth* if for every standard Borel space X and every Borel action of G on X , E_G is smooth (Lean: `AllSmooth`). The Lean definition quantifies over all X in an arbitrary universe u , and the main theorem is universe-polymorphic in u .
- *Nondiscrete* means that the topology of G is not discrete (Lean: `¬ DiscreteTopology G`).

Actions are not assumed continuous: every action whose action map is Borel is covered.

Witness. The Cantor group $C = (\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$ with the product topology, written multiplicatively (Lean: `Cantor := Multiplicative (ℕ -> ZMod 2)`). It is a compact metrizable infinite group, hence Polish and nondiscrete. The witness is not degenerate: it is an uncountable, nondiscrete group, and the property “all-smooth” is not vacuous for it, since it has Borel actions with uncountably many orbits (for instance on C imes C by translation of the first factor), so the smoothness assertion has content. The content of the proof is the general fact that every compact Polish group is all-smooth, which we prove in full (Theorem 1).

2 Proof

Theorem 1 (C2450.allSmooth_of_compact). *Every compact Polish group G is all-smooth.*

Proof. Let G act on a standard Borel space X with Borel action map. Let μ be a (left) Haar measure on G (Mathlib Measure.haar). Since G is compact, μ is finite, and $\mu(G) > 0$ because a Haar measure is positive on nonempty open sets. Let $e : X \rightarrow \mathbb{R}$ be a Borel embedding (Mathlib embeddingReal, a measurable embedding of any standard Borel space into $\mathbb{R}$). For $x \in X$ let $a_x(g) = e(g^{-1} \cdot x)$, a Borel map $G \rightarrow \mathbb{R}$, and define

$$f : X \rightarrow \mathbb{R}^{\mathbb{Q}}, \quad f(x)_q = \mu(a_x^{-1}(-\infty, q)) \quad (q \in \mathbb{Q}).$$

The space $Y = \mathbb{R}^{\mathbb{Q}}$ is Polish (a countable product of Polish spaces).

f is Borel. For fixed q , the set $S_q = \{(x, g) \in X \times G : e(g^{-1} \cdot x) < q\}$ is Borel, and $f(x)_q = \mu(\{g : (x, g) \in S_q\})$ is a Borel function of x by the measurability part of Fubini's theorem (Mathlib measurable_measure_prodMk_left).

f is invariant. For $h \in G$, $g^{-1} \cdot (h \cdot x) = (h^{-1}g)^{-1} \cdot x$, so $a_{h \cdot x}^{-1}(B)$ is the preimage of $a_x^{-1}(B)$ under left multiplication by h^{-1} . Left invariance of μ gives $f(h \cdot x) = f(x)$. Hence $x E_G y$ implies $f(x) = f(y)$.

f separates orbits. Suppose $f(x) = f(y)$ but x and y lie in different orbits. Then the orbits Gx and Gy are disjoint. Fix a Polish topology on X whose Borel sets are the given measurable sets (it exists since X is standard Borel). Each orbit $Gx = \{g \cdot x\}$ is the image of the Polish space G under a Borel map, hence analytic (Mathlib MeasurableSet.analyticSet_image). By the Lusin separation theorem for disjoint analytic sets (Mathlib AnalyticSet.measurablySeparable) there is a Borel set $A \subseteq X$ with $Gx \subseteq A$ and $Gy \cap A = \emptyset$.

Let $\nu_x = (a_x)_* \mu$ and $\nu_y = (a_y)_* \mu$, finite Borel measures on $\mathbb{R}$. The hypothesis $f(x) = f(y)$ says $\nu_x(-\infty, q) = \nu_y(-\infty, q)$ for all rational q (values are finite, so equality of real parts gives equality in $[0, \infty]$), and $\nu_x(\mathbb{R}) = \mu(G) = \nu_y(\mathbb{R})$. The rational half-lines form a π -system generating the Borel sets of $\mathbb{R}$ (Mathlib Real.isPiSystem_Iio_rat, Real.borel_eq_generateFrom_Iio_rat), so $\nu_x = \nu_y$ (Mathlib ext_of_generate_finite). The set $e(A)$ is Borel since e is a measurable embedding. Every $g^{-1} \cdot x$ lies in $Gx \subseteq A$, so $\nu_x(e(A)) = \mu(G)$. If $e(g^{-1} \cdot y) \in e(A)$, injectivity of e gives $g^{-1} \cdot y \in A \cap Gy = \emptyset$; so $\nu_y(e(A)) = 0$. Hence $\mu(G) = 0$, a contradiction. So $f(x) = f(y)$ implies $x E_G y$, and f is a Borel reduction of E_G to equality on the Polish space $\mathbb{R}^{\mathbb{Q}}$. $\square$

Lemma 2 (C2450.cantor_not_discrete). *The Cantor group C is not discrete.*

Proof. C is compact (a product of finite discrete spaces). A compact discrete space is finite (Mathlib finite_of_compact_of_discrete), but $C = (\mathbb{Z}/2)^{\mathbb{N}}$ is infinite. $\square$

Theorem 3 (C2450.exists_nondiscrete_allSmooth_polishGroup). *There exist a Polish group G (with its Borel σ -algebra) that is not discrete and is all-smooth. Witness: $G = C$.*

Proof. C is a compact topological group; it is Polish as a countable product of the finite discrete (hence Polish) spaces $\mathbb{Z}/2$, and its product σ -algebra is its Borel σ -algebra. Apply Theorem 1 and the lemma. $\square$

Remark on the literature. That compact Polish groups have smooth orbit equivalence relations is a classical fact of invariant descriptive set theory. We did not retrieve a source stating it in this session, so we cite none and give the self-contained proof above.

3 Lean formalization

File `lean/Conjecture2450/Basic.lean` (152 lines, only import `Mathlib`, Lean v4.33.1 / `Mathlib` v4.33.1). Definitions: `C2450.IsSmooth`, `C2450.orbitEquivRel`, `C2450.AllSmooth`, `C2450.Cantor`. Theorems: `C2450.allSmooth_of_compact`, `C2450.cantor_not_discrete`, and the main theorem

```
theorem C2450.exists_nondiscrete_allSmooth_polishGroup :  
  exists (G : Type) (_ : Group G) (_ : TopologicalSpace G)  
    (_ : IsTopologicalGroup G) (_ : PolishSpace G)  
    (_ : MeasurableSpace G) (_ : BorelSpace G),  
  Not (DiscreteTopology G) /\ AllSmooth.{u} G
```

where

```
def AllSmooth.{u} (G : Type) [Group G] [MeasurableSpace G] : Prop :=  
  forall (X : Type u) [MeasurableSpace X] [StandardBorelSpace X]  
    [MulAction G X], Measurable (fun p : G x X => p.1 . p.2) ->  
    IsSmooth (orbitEquivRel G X)
```

(ASCII rendering of the Lean source.)

Axiom audit. #print axioms for both `C2450.exists_nondiscrete_allSmooth_polishGroup` and `C2450.allSmooth_of_compact` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide`, and no added axiom. The package builds with `lake build`.